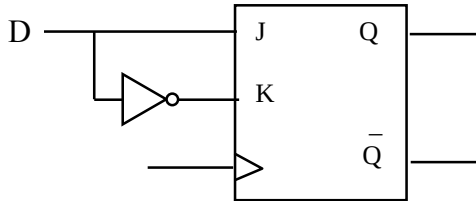


Tutorial 6 (SOLUTIONS)

NOTE THAT THERE ARE OTHER POSSIBLE SOLUTIONS TOO...

Self-Study Question 1

Connect D to J and $\bar{D}$ to K.



Self-Study Question 2

SOLUTION 1

MSOP for $Q^+ = J\bar{Q} + \bar{K}Q$

```
module jkff ( input j, k, clk,
              output reg q);
always @ ( posedge clk )
begin
    q <= ( j & ~q ) | ( ~k & q );
end
endmodule
```

CLK	J	K	Q	Q ⁺
↑	0	0	0	0
↑	0	0	1	1
↑	0	1	0	0
↑	0	1	1	0
↑	1	0	0	1
↑	1	0	1	1
↑	1	1	0	1
↑	1	1	1	0

KMAP for Q ⁺		
Q \ JK	00	01
00	0	1
01	0	0
11	1	0
10	1	1

$$Q^+ = J\bar{Q} + \bar{K}Q$$

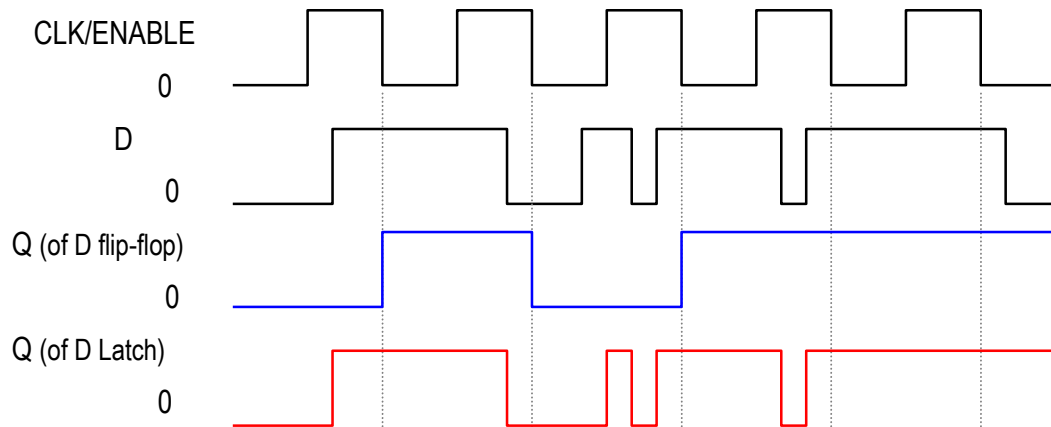
SOLUTION 2 :

```
always @ ( posedge clk )
begin
    q <= j ? ( k ? ~q : 1'b1 ) : ( k ? 1'b0 : q );
end
```

SOLUTION 3 :

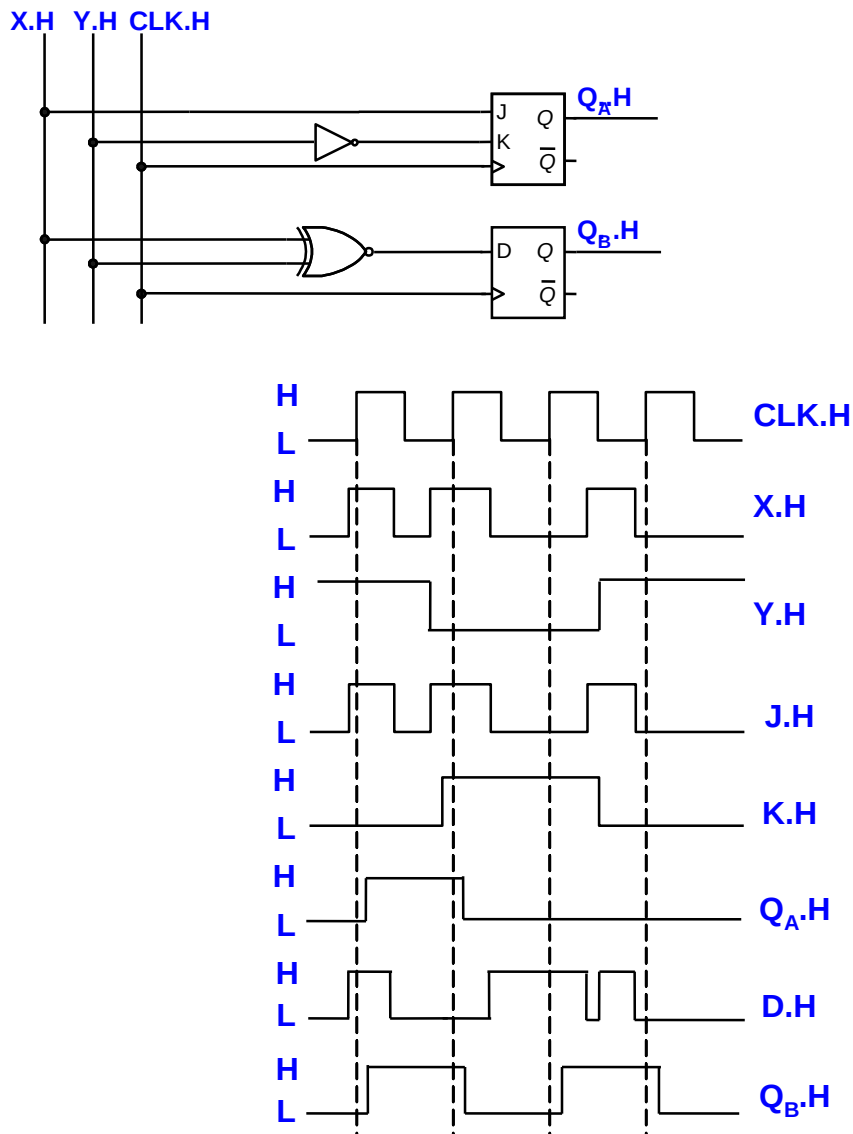
```
always @ ( posedge clk )
begin
    if ( j == 1'b0 && k == 1'b1 )
        q <= 1'b0;
    else if ( j == 1'b1 && k == 1'b0 )
        q <= 1'b1;
    else if ( j == 1'b1 && k == 1'b1 )
        q <= ~q;
    else
        q <= q;
end
```

Question 1

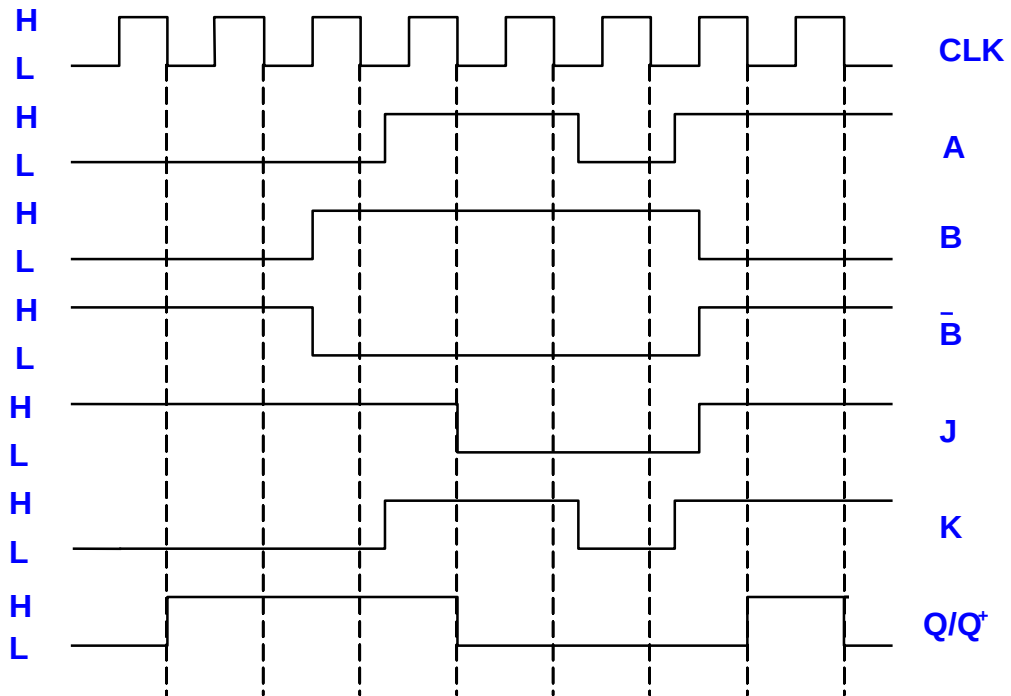
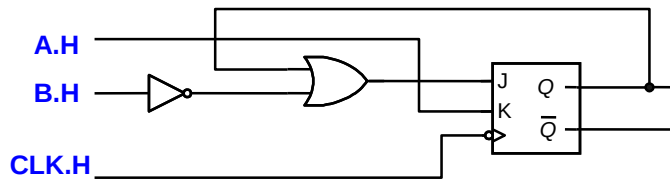


Assumption: The D input satisfies the set-up and hold times of the Latch and flip-flop. Also, the propagation delay of the devices are much less than the CLK period.

Question 2



Question 3



A	B	Q	Q^+
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	0

A	B	Q^+
0	0	1
0	1	Q
1	0	$\bar{Q}$
1	1	0

Condensed Characteristic Table
Characteristic Table